Deepak Mallampati

APPLIED AI ENGINEER - AGENTIC SYSTEMS

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US-based, F-1 OPT, open to relocation

PROFESSIONAL SUMMARY

AI/ML Engineer with 4+ years architecting and shipping agentic AI systems from prototype to production. Built a conductor-subagent orchestration framework that decomposes complex objectives into context-scoped agents (42% token reduction, accuracy sustained), on a FastAPI + Azure OpenAI stack with retrieval and conversational search. Owns technical strategy end to end: reduces per-request cost, improves latency, and scales LLM features to real usage.

CORE COMPETENCIES & SKILLS

Agentic AI & Orchestration • Multi-Agent Systems • FastAPI • Azure OpenAI • RAG & Semantic Search • Prototype to Production • Cost & Latency Optimization • LangChain / LangGraph • LLM Evaluation

Generative AI & LLMs: GPT-4o, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | RAG Architecture | Multi-Agent Systems | Fine-tuning (LoRA, QLoRA, PEFT) | Hugging Face Transformers

ML & NLP: PyTorch, TensorFlow, Keras | NER, Text Classification, Summarization | Model Optimization | RAGAS Evaluation | BERTScore | Drift Detection

Vector Search & Retrieval: FAISS, Pinecone, Weaviate | Semantic & Hybrid Search | Cross-Encoder Re-ranking | Semantic Caching

MLOps & Cloud: AWS (Bedrock, SageMaker), Azure OpenAI, GCP Vertex AI | Docker, Kubernetes | Terraform | CI/CD | MLflow, Weights & Biases

Data & Compliance: PostgreSQL, MongoDB, Redis | Spark, Kafka | HIPAA, SOC 2, PCI DSS | PII Masking | Prompt Injection Mitigation

Languages & Tools: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | Git, Jira

PROFESSIONAL EXPERIENCE

Progress Solutions *Jul 2025 - Present*

AI Engineer

USA

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Built a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency and accuracy dashboards to benchmark model iterations pre-deployment and surface regressions before release.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.

Vanguard *Jan 2024 - Jan 2025*

AI Engineer Intern

USA

- Developed **AI agents and backend services** for Vanguard's internal AI platform across **25+ AI agents** and workflows, integrating with production data pipelines and observability tooling.
- Optimized LLM prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** and informing platform model-selection decisions.
- Built **12 APIs and microservices** powering new AI capabilities, cutting p95 latency from 1.5s to **800ms** and supporting **100,000+ requests daily** across 20+ internal teams.
- Engineered content controls for secure, policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining response quality.

Kent State University *Oct 2023 - Jan 2024*

ML & Gen AI Research Assistant

USA

- Built a **privacy-preserving ML pipeline** on a clinical hospital-readmission dataset, applying k-anonymity, l-diversity, and Laplace differential privacy across 8 clinical features.
- Reduced record-linkage re-identification risk from **3.38% to 0.32%** while retaining **99% of baseline predictive performance** (Random Forest AUC 0.689 vs. 0.696).
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 with a curated instruction dataset for controllable long-form generation.

Emerson *Jun 2022 - Apr 2023*

ML Trainee

India

- Trained supervised **regression and classification models** on operational pipeline-storage data for equipment-failure prediction, anomaly detection, and capacity forecasting.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification on domain-specific datasets, achieving **89% F1-score** on custom NER tasks.

PROJECTS

MangaLens Chrome Web Store

Shipped Chrome extension delivering real-time AI translation of manga and webtoons across 29+ sites, turning a multi-step manual translation chore into a single inline action.

Privacy-Preserving Clinical ML Pipeline

Framework combining k-anonymity, l-diversity, and differential privacy so researchers can analyze sensitive patient data without exposing identities, quantifying the privacy-utility trade-off precisely.

Story Director LLM

Fine-tuned language model that generates structured short-form video narratives from a single prompt, paired with an automated rendering pipeline.